User's Manual
PVMD Toolbox

Developed by:
Photovoltaic Materials and Devices
Technical University of Delft

User's Manual PVMD Toolbox

by

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1. Introduction

The PVMD TOOLBOX is a MATLAB-based software package developed by the Photovoltaic Materials & Devices Group (PVMD Group) at the Delft University of Technology, Delft, the Netherlands. It is a sophisticated tool that allows users to model the performance of solar photovoltaic systems with great detail and accuracy. The latest update to the toolbox comes with additional features to broaden its capabilities. This document is a starters manual that explains the basic principles of the PVMD Toolbox and how it can be used.

Within this chapter, Section 1.1 explains how the Toolbox can be installed. Section 1.2 depicts the workflow adopted to perform the calculations. Section 1.3 explains how an certain input while can be selected, while Section 1.4 explains how simulationed results can be saved.

The toolbox includes the following simulation steps:

(A) CELL - Compute and evaluate the performance of interested solar cell chemistry.

(B) MODULE - Calculate and visualize the optical performance of a defined PV system.

(C) WEATHER - Determine the amount of energy absorbed by the solar cells before electrical conversion.

(D) THERMAL - Calculate the operational temperatures of the PV system.

(E) ELECTRIC - Compute the DC Energy output from the defined PV system.

(F) CONVERSION - Account for the conversion losses to obtain the generated AC electricity.

The steps are performed in the order specified in the aforementioned list whilst running the toolbox. Moreover, the different possibilities of computations within the toolbox and the nature of the corresponding output variables will be defined in the later chapters of this user manual.

1.1 How to install the PVMD Toolbox

If the zip-file has correctly been received or downloaded, it should contain an executable named "PVMD_TOOLBOX_app". This executable should be run, such that the installation will start. Once the installation is complete, the PVMD Toolbox should be a program on your laptop or computer.

If this program is opened, it shows the home screen of the Graphical User Interface (GUI), as shown in Figure 1.1. In top of this window, there is a tab for each of the earlier-mentioned simulation steps of the PVMD Toolbox. More information on the workflow is available in the next section. From this starting screen, it is possible to access the website with all information and explanation on the PVMD Toolbox. Also, an earlier used simulation can be loaded from this screen and the simulated results can be saved. Both of these options will be explained later in this chapter.

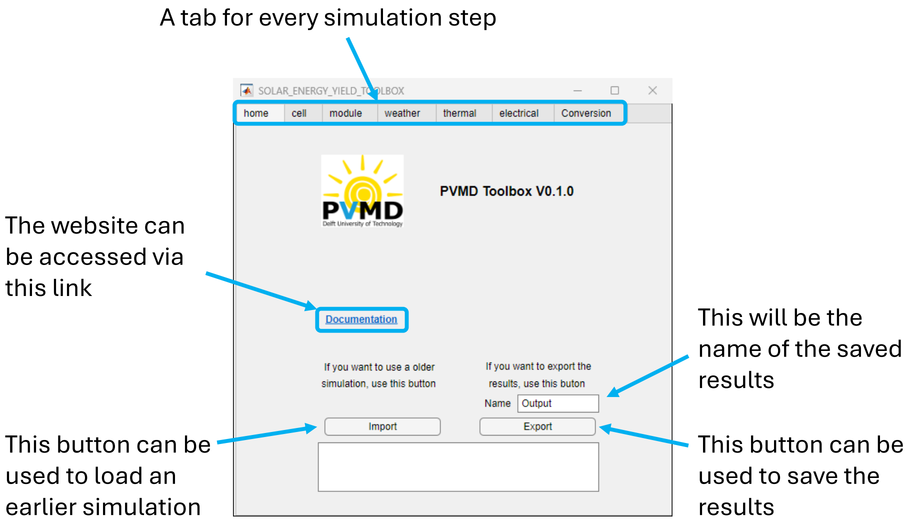

Figure 1.1: The homescreen of the PVMD Toolbox

1.2 Toolbox workflow

In this section, the general workflow diagram on how data flows between the various sections of the toolbox is presented. The workflow diagram associated with this version of the TOOLBOX has been provided in Figure 1.2. In this figure:

- The blue rectangles represent the different blocks of the Toolbox. Each block is based on one or multiple physical models, which is described in more detail in the individual chapters.
- The black lines represent the different inputs that are needed at the different stages of the PVMD Toolbox.
- The green lines represent the different outputs of the Toolbox.

It should be mentioned that each block requires the input from the previous block. Therefore, all blocks should be run in sequential order. Additionally, if a certain simulation step is redone with different input values, this will not directly affect the simulated results of the steps after

1.3 Load an earlier simulation

If an earlier simulation has been saved, it is possible to load this simulation and use it as starting point. All the choices and simulated results from this simulations will be loaded and inserted in the corresponding tab. This can be done with the button in the bottom left corner, as indicated in Figure 1.1. If this button is clicked, a window will appear that shows your directory. From this directory, a file can be selected that contains the desired simulations. It should be noted that only files ending in '.mat' can be opened.

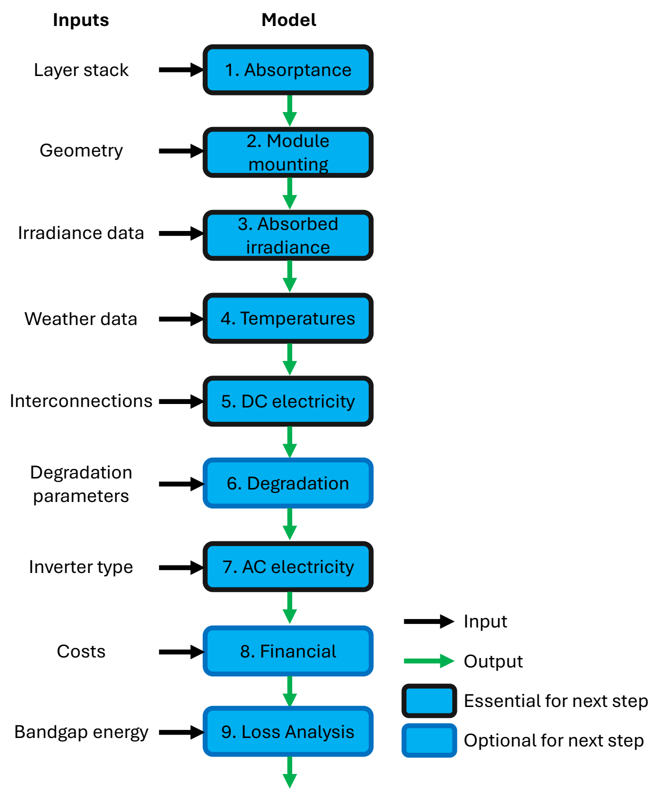

Figure 1.2: PVMD Toolbox Workflow Diagram

1.4 Saving an output

Once all the desired simulation steps have been completed or the made progress needs to be used in a later simulation, it is possible to save the results and selected inputs via the button in the bottom right of the home screen, as indicated in Figure 1.1. The filename of the saved file can be chosen in the textbox above the bottom. Once the button is pressed, a window will appear showing the directory. This allows the user to choose the location where the results should be saved.

2. Cell

If the cell-tab is selected, the user will see either one of the screens shown in Figure 2.1. One can alternate between the screen by using the drop-down menu in the top left corner of the screen. This drop-down menu has the option to "Run GenPro" or to "Load Simulation".

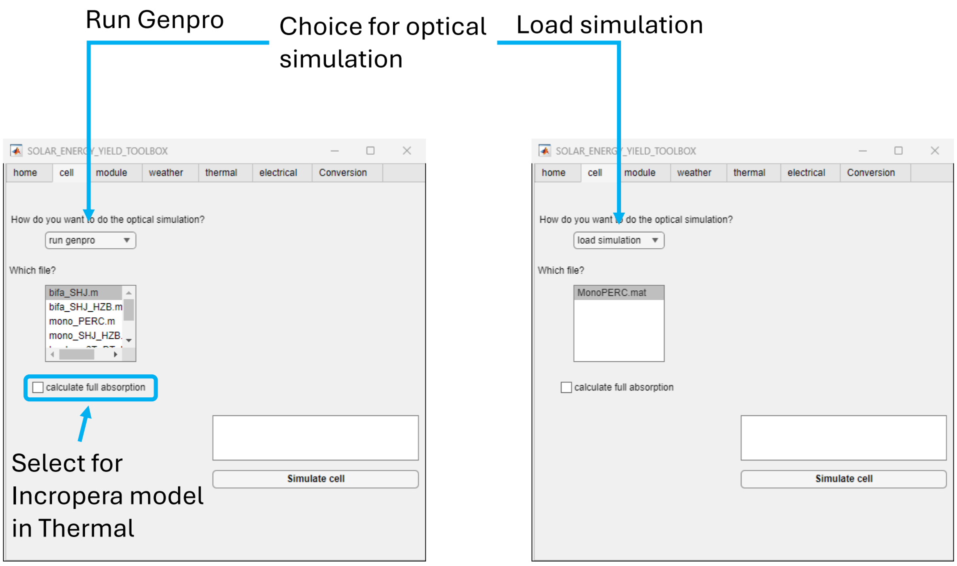

Figure 2.1: The window for the cell simulation

2.1 Run GenPro

If the option "Run GenPro" is selected, the textbox below it will show all possible cell structures that can be simulated with the in-house developed GenPro. More information on the working principles and operation of GenPro is provided in the extensive Toolbox explanation. When the desired cell structure is selected, the button "Simulate cell" on the bottom right can be pushed, starting the simulation with GenPro. This simulation will calculate the absorptance profile of each absorber layer of the cell.

Besides the selection of the cell structure, it is also possible to check a box labelled "Calculate full absorption". When this box is checked, the absorptance is not only calculated in the absorber layers, but in all layers in the cell stack. This is only needed when the Incropera model is used in the Thermal part. In all other situations, this check box can be checked off.

2.2 Load simulation

If the option "Load Simulation" is selected, it is possible to select a reference/ example simulation. As the simulation of GenPro can be computationally intensive, it can be desired to skip this step and use a simulation from the library.

2.3 Simulated results

Once the Cell simulation has been completed, a figure will appear in the top right of the window. An example of this figure is shown in Figure 2.2. This output contains the absorptance profile of all absorber layers in the cell over the full wavelength range. Although this absorptance profile is calculated for different angles of incidences, only the one for the first angle is plotted. The profiles of the other angles are saved as output such that they can be used for other simulation steps.

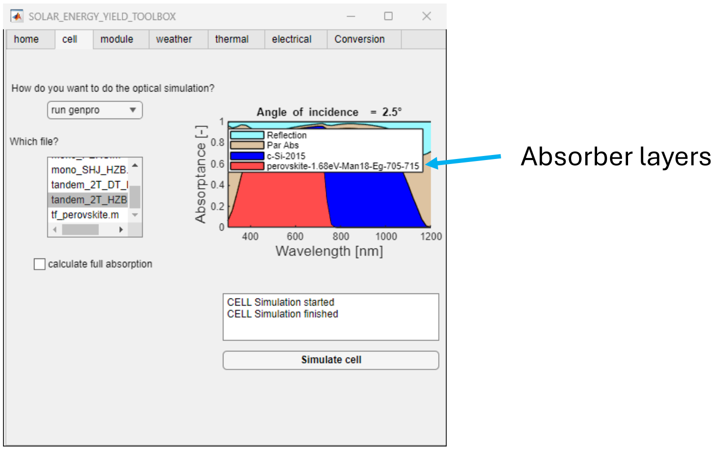

Figure 2.2: The result of the cell simulation

3. Module

Once the Cell simulations are completed and the Module-tab is selected, the user will see the screen as shown in Figure 3.1. This step will produce a so called sensitivity map, representing the optical response of the PV module to different sky segments. The full explanation of the sensitivity map is provided in the extensive Toolbox explanation. Similar to the Cell simulation, there is a drop down menu that allows the user to select the type of simulation that it wants to perform. In the current status of the PVMD Toolbox, the options "Periodic", "Non-Periodic", and "Tracking" are possible.

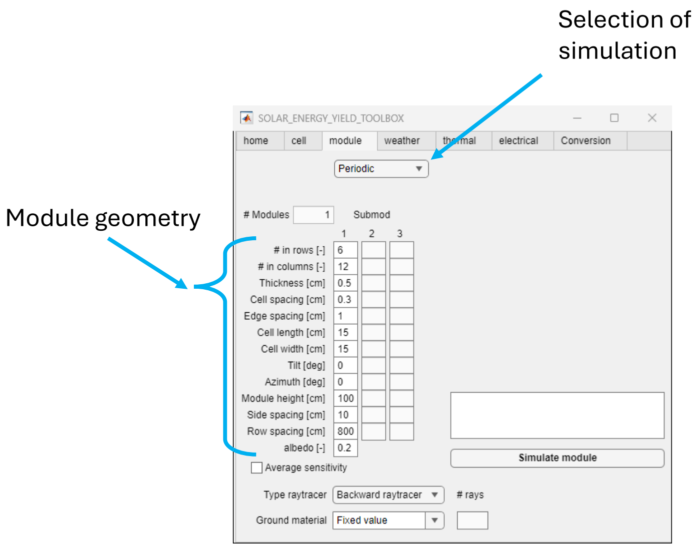

Figure 3.1: The window for the module simulation

3.1 Periodic

The periodic simulation should be selected if the case study considers a typical fixed PV module operating in the field. For a periodic simulation, no other objects than PV modules and the ground are considered and only shading from other PV modules is included. Also a PV module in the middle of the field will be simulated, ignoring effects present at the edges of a PV farm. As only one module is considered, the textbox labelled "# Modules" cannot be changed. Changing this box is only possible in the Non-Periodic simulation.

The periodic simulation requires different inputs from the user regarding the geometry of the module. All required geometry inputs, including their meaning and valid inputs are listed in Table

It can be seen that there are multiple columns for all geometry inputs. These columns can be used in the PV module is composed of different submodules, such as in four-terminal configurations. In this case, the cell layout of the different submodules can be different. In case of two-terminal configurations, only the first column can be changed.

Below all geometry inputs, there is also a checkbox labelled "Average sensitivity". By checking this box, the sensivity over all cells is averaged, meaning all cells will receive the same irradiance. This can be useful when it is not desired to have deviations among the cells and it can speed up the ray tracing process for the backward ray tracer.

Label	Meaning	Valid inputs
# in rows [-]	Number of cells in a row of the module.	Positive integer
# in columns [-]	Number of cells in a column of the module.	Positive integer
Thickness [cm]	The thickness of the PV module.	Positive number
Cell spacing [cm]	The spacing between cells within the module.	Positive number
Edge spacing [cm]	The spacing between cells and the edge of the module.	Positive number
Cell length [cm]	The length of the cells.	Positive number
Cell width [cm]	The width of the cells.	Positive number
Tilt [$\circ$]	The tilt of the PV module (0 refers to completely flat, 90 to vertical).	Number between 0 and 90
Azimuth [$\circ$]	The azimuth of the PV module (0 refers to south, 90 to west, 180 to north, 270 to east).	Number between 0 and 360
Module height [cm]	The height of the PV module with respect to the ground.	Positive number
Side spacing [cm]	The distance between the modules on the left and right sides.	Positive number
Row spacing [cm]	The distance between the modules on the front and rear sides.	Positive number
Albedo [-]	The effective albedo of the ground. Note that this input is only used when the selected ground material is "Fixed Value".	Number between 0 and 1

Table 3.1: The different geometry inputs that need to be provided.

Another important choice that can be made in the periodic simulation is the type of raytracer that is used. Either the "Backward raytracer" or "Forward raytracer" can be selected. The differences between these options is discussed in full detail in the extensive Toolbox explanation. If the "Forward raytracer" is selected, the number of rays should also be filled in in the corresponding textbox (bottom middle). A typical value for the number of rays is 500, and a higher value will reduce the uncertainty in the simulation but also increases the computational time.

The last choice that can be made in the Periodic simulation is the ground material. This selects which spectral reflection profile should be taken for the ground. The list in the drop-down menu shows all materials that are included in the library. It is also possible to use a fixed value for the ground reflections over the wavlength range, by selecting the option "Fixed Value". In this case, the albedo value from Table 3.1 is taken.

3.2 Non-Periodic

The Non-Periodic simulation should be selected when only a few modules in a specific environment are considered. Once this option is selected, the window in Figure 3.2 will be shown. The environment can be selected with the drop-down menu in the bottom left. The procedure for adding an environment to the library is discussed in Appendix.

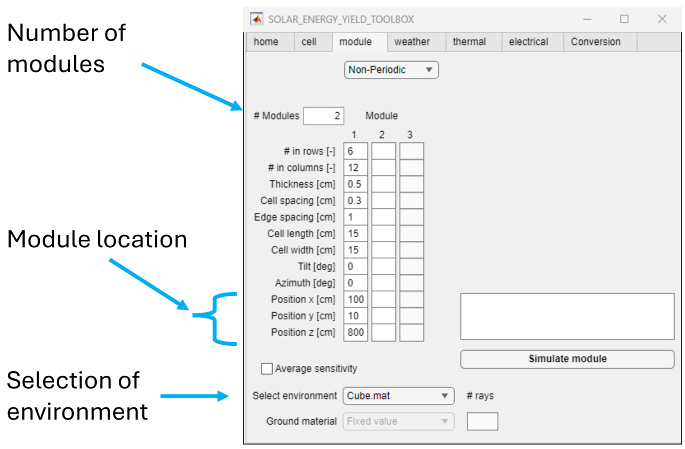

Figure 3.2: The window for the Non-Periodic module simulation

In contrast to the periodic simulation, it is now possible to add multiple modules (to a maximum of 3 in the GUI version) with the textfield in the top left. One can see that by changing the inserted number, different columns become active. Especially, the bottom three rows are important as they describe the position of the bottom left corner of the PV module in the scenario.

3.3 Tracking

The tracking option allows the user to select dynamic modules that can follow the sun. Once this option is selected, the screen in Figure 3.3 will be shown. Three different tracking options are possible:

- Horizontal Single Axis Tracking Systems (HSATS)
- Tilted Single Axis Tracking Systems (TSATS)
- Double Axis Tracking Systems (DATS)

Because the module orientation will vary during its operation, no module tilt or azimuth can be assigned to the system. Therefore, these fields are disabled when a tracking simulation is chosen. The orientation of the tracking system can be changed by the adjusting the advanced settings, as explained in Appendix A.

One thing that is different for the tracking simulation in contrast to the periodic and the non-periodic simulation, is that the WEATHER simulation is already carried out in this stage. Therefore, the inputs for the WEATHER should also be specified at the next tab. Then the 'Simulate Module' button can be pushed and the weather simulation does not need to be run separately.

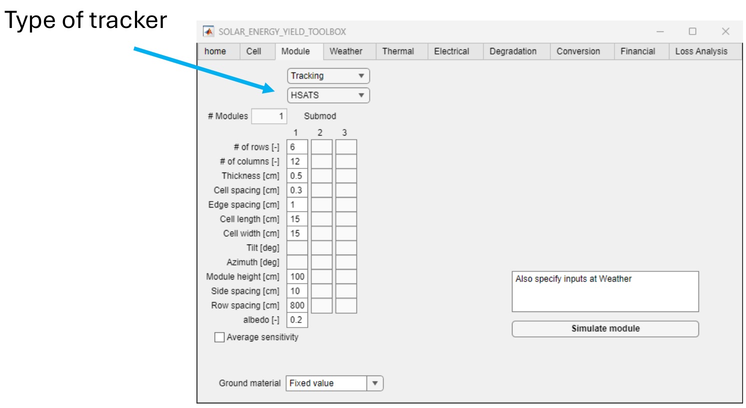

Figure 3.3: The window for the Tracking module simulation

3.4 Simulated results

As mentioned before, the Module step generated a sensitivity map that indicates the optical response of the PV module to the different sky segments. After running this step, an average value of this sensitivity map will be displayed in the GUI, as shown in Figure 3.4. It can be seen that the sensitivity map in the figure is more sensitive to the sky segments in the south, as it has a tilt of 30° and is facing south (azimuth is 0°).

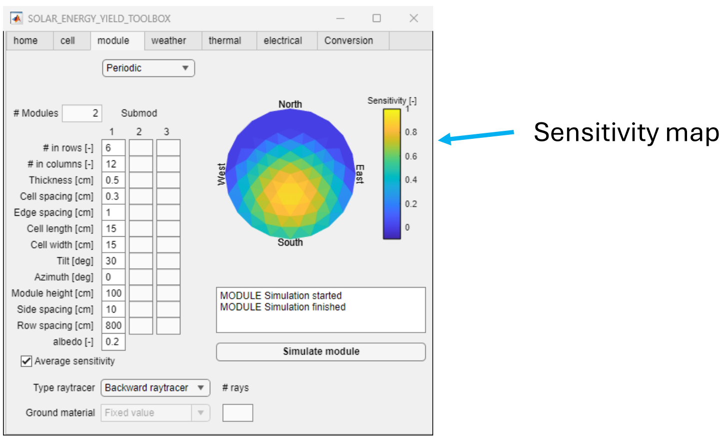

Figure 3.4: The result of the module simulation

4. Weather

After the module simulation, the irradiance can be calculated in the weather step, which contains the screen shown in Figure 4.1.

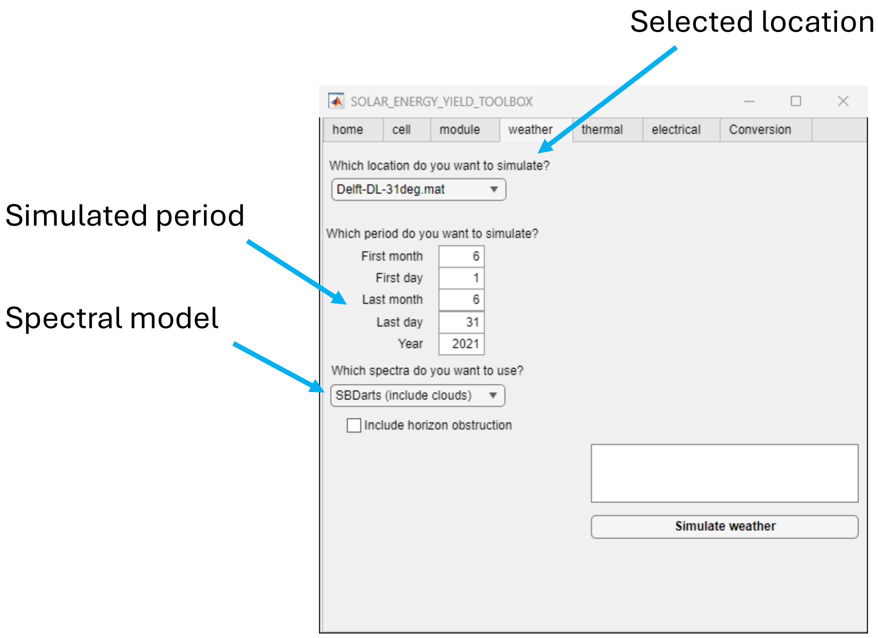

Figure 4.1: The window for the weather simulation

4.1 General choices

The most important choice is the location for which the simulation is performed, which can be chosen by the drop-down menu in the top left. Other than the selected location, also the simulation period is relevant. In the text boxes in the left middle, it is possible to indicate from when till when the simulation should be performed.

The next choice that can be made is which model is used for the spectral calculation. Either the "SMART" or "SBDarts" model can be selected, where the main difference is that the first model does not consider cloud effects, which is the case for the later one. The full details on how these models are used is explained in the extensive Toolbox explanation. In most cases, the SBDarts model is recommended as it has demonstrated a higher accuracy than SMART.

4.2 Account for horizon obstruction

Below the drop-down menu of the spectral model, there is a checkbox that enables the user to account for an horizon obstruction. This option should only be selected if there objects or buildings should be considered that are not included in the module simulations. In other cases, a free-horizon can be used and the checkbox should be kept empty.

If the option is selected, a new drop-down menu will appear, as shown in Figure 4.2. The menu contains all horizon obstructions saved in the library. The procedure for adding a new horizon obstruction to the library is discussed in Appendix.

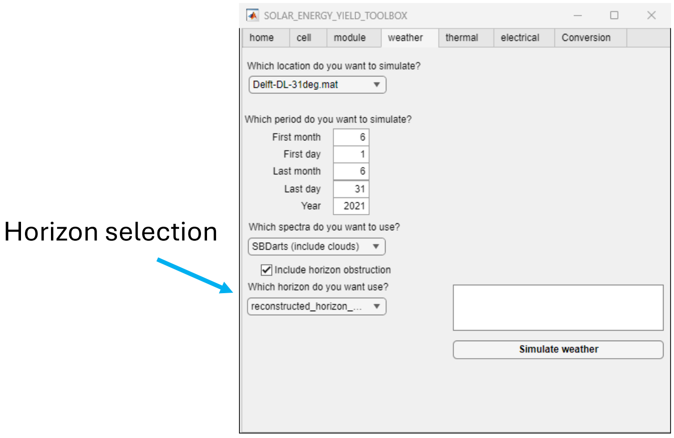

Figure 4.2: The selection for the horizon obstruction

4.3 Simulated results

Once the weather simulation is finished, the figure of the daily irradiation will be displayed, as shown in Figure 4.3. Besides the irradiation, also the absorbed photo-current is calculated, which will be used as input for the next simulation steps (as shown in Figure 1.2).

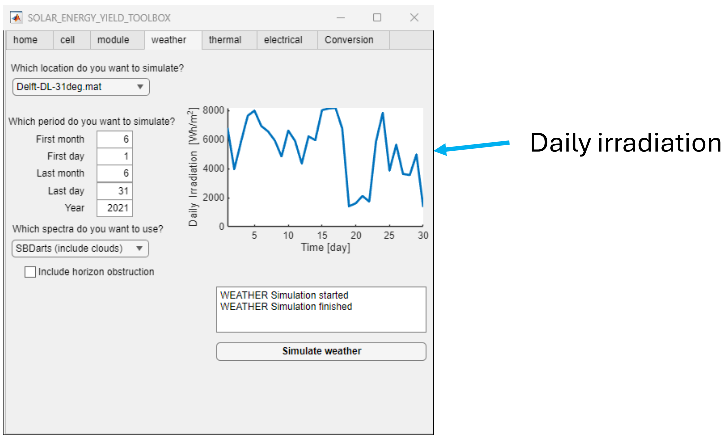

Figure 4.3: The result of the weather simulation

5. Thermal

The absorbed irradiance is used to calculate the temperature of each cell through the simulated period. Once the thermal tab is selected, the window in Figure 5.1.

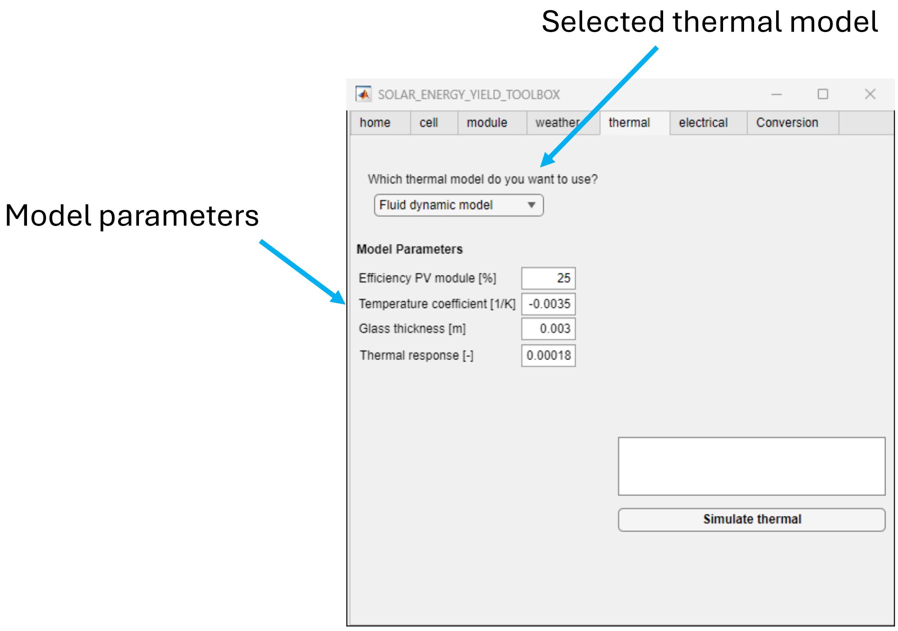

Figure 5.1: The window for the thermal simulation

5.1 General choices

It is possible to use different models for the thermal calculation. The drop-down menu in top left corner shows all available thermal models, which include

- the Fluid dynamic model
- the Duffie-Beckman model
- the Faiman model
- the Sandia model

An explanation and comparison of all thermal models is provided in the extensive Toolbox explanation. For each model, certain parameters are used to perform the simulation. By changing the selected thermal model, the required parameters will also vary. A detailed explanation on the parameters and their typical values for each model can also be found in the extensive Toolbox explanation. For most cases, the fluid dynamic model is recommended to be used, as it is not based on empirical parameters.

5.2 Simulated results

Once the thermal simulation is finished, the window will show the range of cell temperatures throughout the simulation period, as shown in Figure 4.3. These temperatures will be used as input for the electrical simulation in the next step.

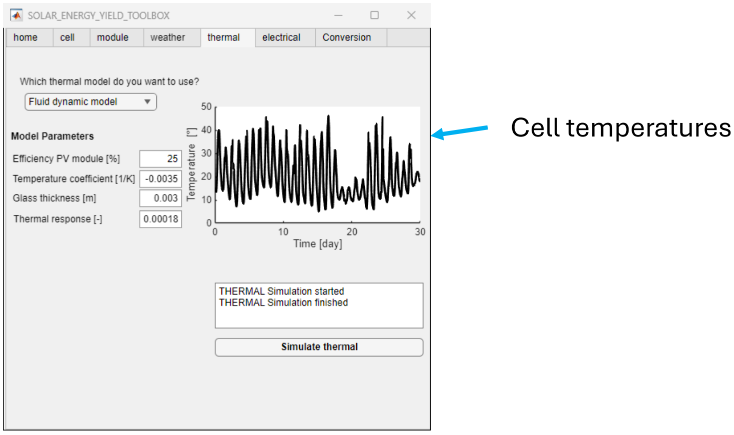

Figure 5.2: The result of the thermal simulation

6. Electrical

Once the irradiance and cell temperature are calculated, the electrical performance can be simulated. Once the thermal tab is selected, the window in Figure 6.1 will be visible.

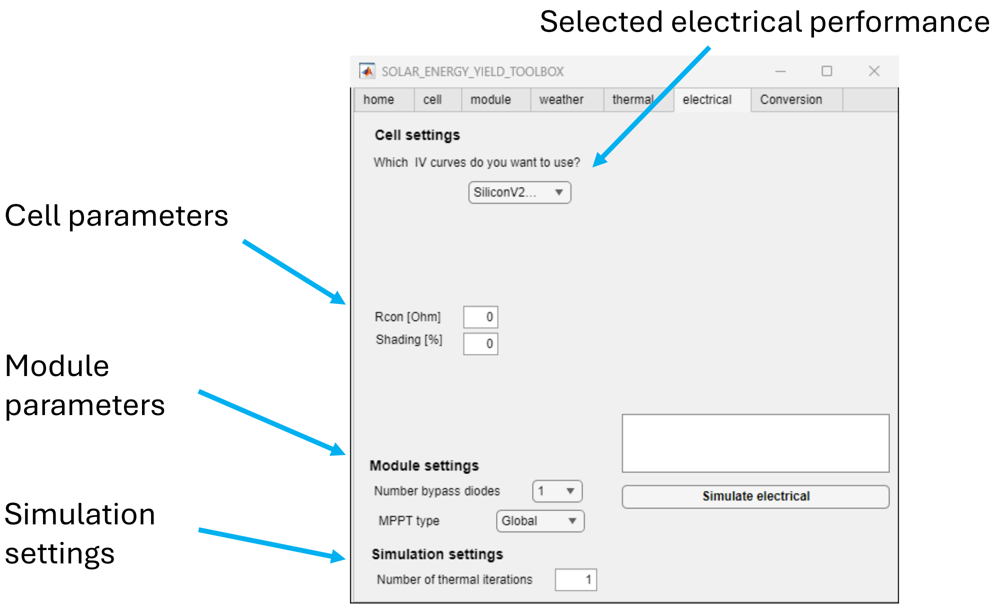

Figure 6.1: The window for the electrical simulation

6.1 General choices

The first choice that needs to be made is the electrical performance of the solar cells. This includes the temperature- and illumination-dependent forward bias performance, the reverse bias performance, as the performance for light soaking. The explanation on all these aspects is provided in the extensive Toolbox explanation. The drop-down menu in the top left corner lists all available cell technology from the library. The procedure for adding a new option to the list can be found in Appendix.

Besides the cell technology, also the contact resistance and shading should be provided. The contact resistance accounts for all ohmic losses for the transport and collection of current through the cells. The shading loss accounts for the loss in active area due to the shading of the contacts. It should be noted that when these losses are included in the selected electrical performance-model, both values can be set to 0.

Below the cell parameters, there are also module parameters that should be selected. This include the number of by-pass diodes and the type of maximum power point tracker.

6.2 Multi-junction devices

If the simulated cell structure in step 1 is a multi-junction cell, more inputs will be visible, as shown in Figure 6.2. First of all, for each subcell a choice should be made for the electrical performance. Additionally, multi-junction devices have the so-called *photon recycling effect*, where recombined photons can be reabsorbed in other subcells. The full procedure of this is explained in the extensive Toolbox explanation. To account for this effect, the luminescence efficiency should be specified. If this effect does not need to be considered, its value can be set to 0.

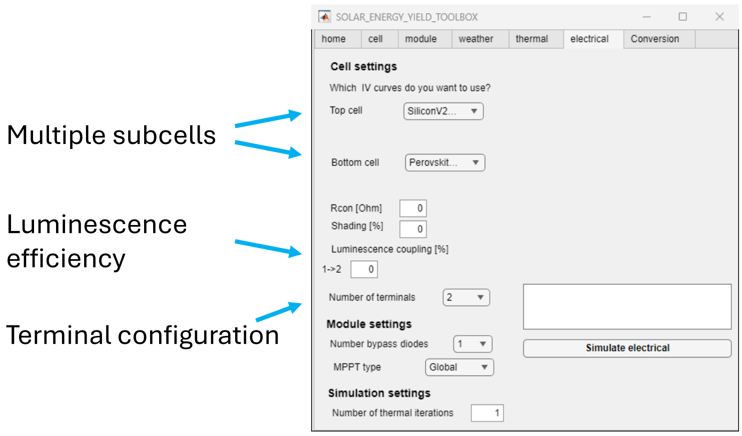

Figure 6.2: The window when a multi-junction solar cell is considered

The last choice that needs to be made for multi-junction devices is the terminal configuration. In the drop-down menu, it can be selected whether the simulated cells are two-, three-, or four-terminal connected. A deep explanation on the differences between these configurations can be found in the extensive Toolbox simulation.

6.3 Simulated results

Once the electrical simulation is finished, the window will show the daily generated electricity, as illustrated in Figure 6.3. Besides the electricity generation, also outputs are exported needed to reconstruct the IV curves at any time.

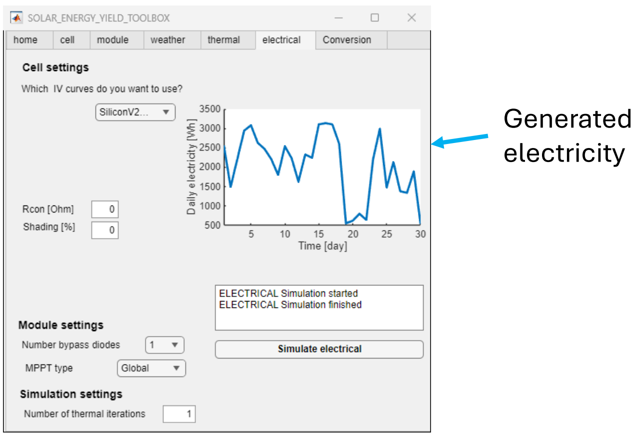

Figure 6.3: The result of the electrical simulation

7. DEGRADATION

Besides the energy yield performance at the initial year, it is also possible to calculate the energy yield after the specified years of degradation. Different degradation mechanisms are included, such as moisture induced degradation, discoloration, thermal cycling and light induced degradation. Additionally, a specified degradation rate for the perovskite cell can be specified in the case of multi-junction devices. Once the Degradation tab is selected, the window of Figure 7.1 will be visible.

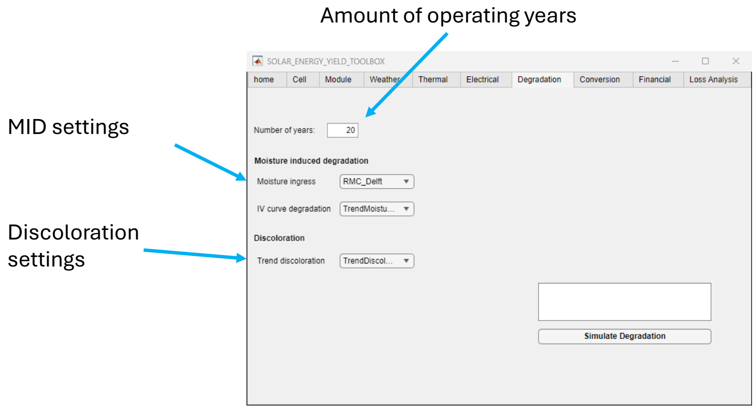

Figure 7.1: The window for the electrical simulation

7.1 General choices

The user needs to specify different inputs for the simulation of the degradation. First, the amount of operating years that should be considered has to be selected. This will be used to determine for how many degradation processes should be simulated. It should be noted that the output power and energy yield is only calculated for the selected number of years, for a time-resolved energy yield, the model needs to be run multiple times.

Then, regarding the moisture induced degradation, the simulation file of the relative moisture ingress has be be selected, and the trend in IV curve degradation has to be chosen. The relative moisture ingress contains the simulated (or measured) results of moisture ingress into a PV module for a certain location. It should be realized that the location of the moisture simulation should match the location of the PV system. The trend in IV curve contains a description of how the equivalent circuit parameters are affected by the presence of moisture inside the module.

Lastly, for the discoloration, the trend of change in refractive index has to be specified. The selected file is used to update the refractive index of the encapsulant according to the amount of received irradiance and temperature.

7.2 Multi-junction devices

When multi-junction devices are considered, a degradation rate for the perovskite cell should be specified. Also, the degradation scenario (i.e. loss in voltage, current or fill factor) should be selected. The corresponding display is shown in Figure 7.2.

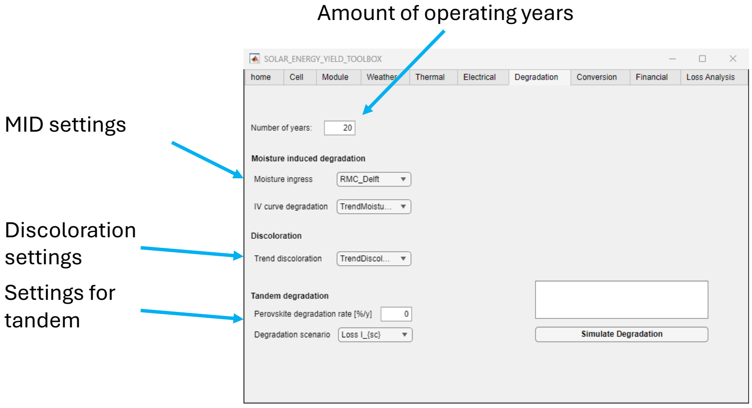

Figure 7.2: The window when a multi-junction solar cell is considered

7.3 Simulated results

The PVMD Toolbox will calculate the degradation rates of the different mechanisms and the electrical performance of the degraded module after the specified years of operation. The GUI will only plot the degradation rates, but the electrical performance is present when the output is exported.

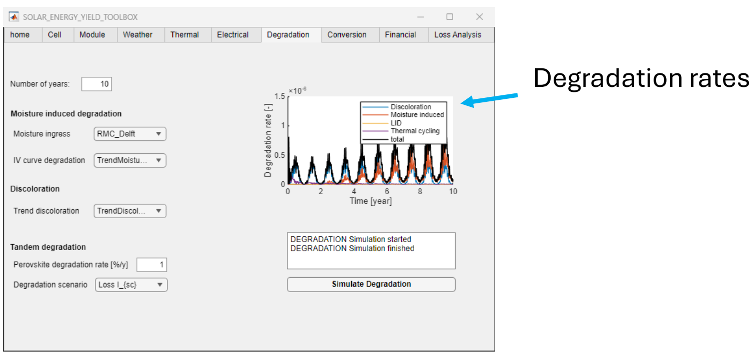

Figure 7.3: The result of the degradation simulation

8. CONVERSION

After the DC power calculation, the inverter losses can be considered. Once the tab 'Conversion' is selected, the window of Figure 8.1 will be visible.

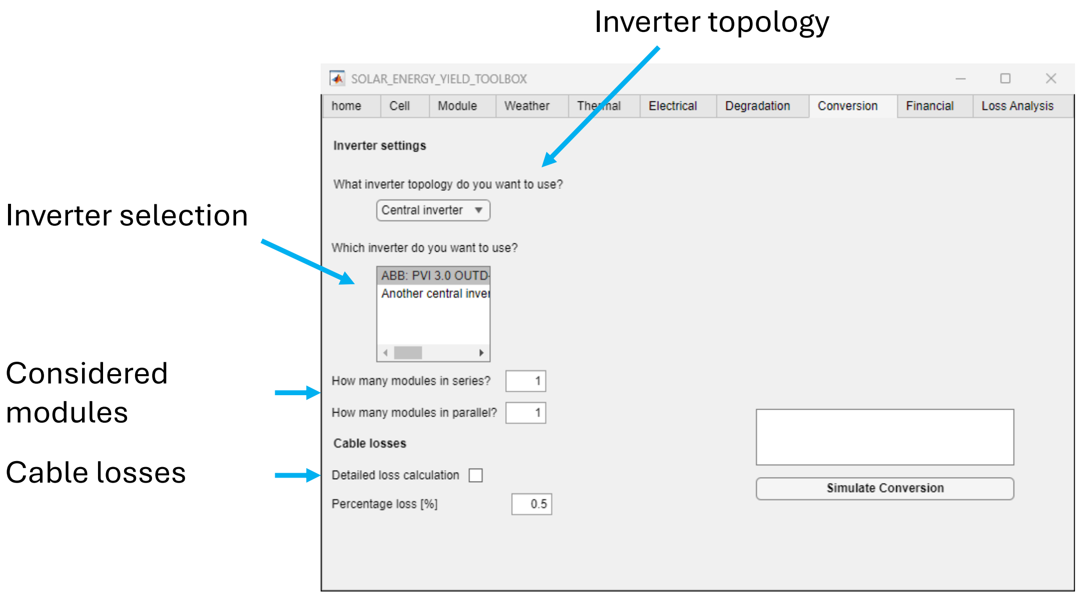

Figure 8.1: The window for the conversion simulation

8.1 General choices

The most important choice is the converter topology, which is selected in the first drop-down box. The selected option can slightly adjust the other input questions, such as the number of modules in series or parallel.

After the inverter topology, the inverter should be selected. Besides the options in the display, it is also possible to select an a different inverter.

For the cabling losses, either a fixed loss percentage can be considered (when the box is not checked) or a detailed calculation can be performed (when the box is checked). When the box is checked, different options will appear to indicate the length, area, and resistivity of the cables.

8.2 Results

Once the simulation is finished, the figure will show the AC output power throughout the simulated period.

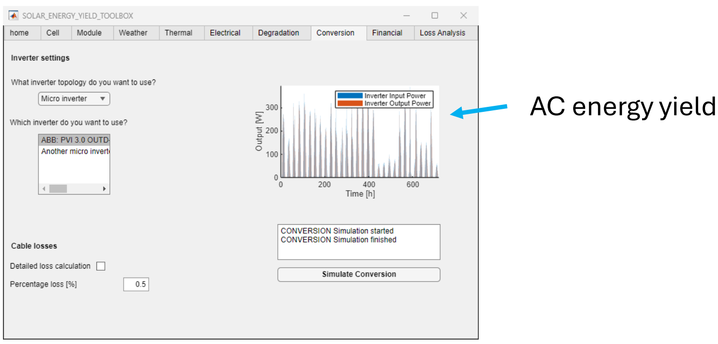

Figure 8.2: The results for the conversion simulation

9. FINANCIAL

At the moment, the financial part is not working properly. The GUI and manual will be updated as soon as possible with a version that includes financial calculations.

10. LOSS ANALYSIS

As a last analysis, the loss distribution of the PV system can be calculated Once the tab 'LOSS ANALYSIS' is selected, the window of Figure will be visible.

10.1 General choices

10.2 Results

A. Advanced Settings

Besides the choices that should be provided in the GUI, it is possible to change more detailed settings. This should be done by changing (or recreating) the excel file 'Settings.xlsx'. This file contains different tabs that contain settings about different parts of the PVMD Toolbox. It is recommended to only change settings when the user is aware of what certain values mean. The default value is recommended otherwise.

A.1 Uploading a new settings document

B. Create a new cell structure

C. Create a new IV characteristic file